Bellabeat Case Study

How Can a Wellness Company Play It Smart?

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1. Introduction

Bellabeat is a high-tech wellness company that manufactures health-focused smart products for women. The purpose of this case study is to analyze publicly available Fitbit fitness tracker data to identify consumer usage trends and develop high-level marketing recommendations for one Bellabeat product.

2. Ask

Business Task

Analyze smart device usage data to identify trends in consumer behavior and apply those insights to one Bellabeat product in order to develop high-level marketing recommendations that support Bellabeat's marketing strategy.

Problem Statement

Bellabeat seeks to better understand how consumers use non-Bellabeat smart devices. By examining activity, sleep, and wellness data, the company aims to uncover meaningful patterns that can be used to improve marketing efforts and encourage adoption of Bellabeat products.

Guiding Question 1: What is the problem you are trying to solve?

The problem is that Bellabeat needs evidence-based insight into consumer smart device usage. Without analyzing user data, the company cannot confidently identify customer habits or determine which marketing strategies are most likely to resonate with potential customers.

Guiding Question 2: How can your insights drive business decisions?

The insights generated from this analysis can help Bellabeat identify common wellness behaviors, recognize opportunities for targeted marketing, improve product positioning, and support strategic decision-making by the marketing analytics team and executive leadership.

Key Stakeholders

- Urška Sršen – Co-founder and Chief Creative Officer
- Sando Mur – Co-founder and Executive Team Member
- Bellabeat Marketing Analytics Team
- Bellabeat Executive Team

Deliverable

The business task for this project is to analyze smart device usage data to identify trends in how consumers use non-Bellabeat smart devices. The insights gained from this analysis will be applied to a selected Bellabeat product to develop high-level, data-driven marketing recommendations that support Bellabeat's business objectives and future growth.

3. Prepare

Data Sources

This analysis uses one primary data source: the **FitBit Fitness Tracker Data** dataset, which is publicly available through Kaggle and provided by **Mobius** under the **CC0: Public Domain** license. The dataset is described as containing fitness tracker data from approximately **30 Fitbit users** who voluntarily consented to share their personal activity information. However, validation of the raw files in **Google BigQuery** showed that the number of distinct participant IDs varies across tables and collection periods, ranging from **24 to 35 participants** in the datasets used for this analysis. The data includes daily physical activity, step counts, heart rate, sleep monitoring, and other wellness-related metrics. No additional datasets were used for this analysis.

Direct link: FitBit Fitness Tracker Data

Why This Dataset Was Selected

The Fitbit Fitness Tracker dataset was selected because it directly supports the objectives of this case study. Bellabeat's goal is to understand how consumers use smart fitness devices in order to develop data-driven marketing strategies. Although the dataset contains information from Fitbit users rather than Bellabeat customers, it provides valuable insights into user activity, sleep habits, and overall wellness behaviors that can be applied to Bellabeat's products and marketing initiatives.

Data Organization

The Fitbit dataset is organized into multiple CSV files, with each file representing a different category of fitness tracker information, such as daily activity, sleep, heart rate, hourly activity, and weight logs. Each record is associated with a unique user identifier (Id), allowing related datasets to be connected during analysis when appropriate. The case study notes that the starter analysis commonly focuses on the **dailyActivity_merged** and **sleepDay_merged** datasets, although additional files may be incorporated depending on the scope of the analysis.

Data Credibility (ROCCC Evaluation)

The dataset was evaluated using Google's **ROCCC** framework to determine its suitability for analysis.

Reliable:
The dataset provides structured measurements of user activity and wellness behaviors collected through Fitbit devices. However, its reliability is limited by the relatively small sample size, inconsistent participant coverage across tables, and limited information regarding the original participant recruitment and data-collection methodology. Therefore, the dataset is appropriate for

exploratory analysis, but the findings should not be generalized to the broader population without additional supporting data.

Original:
The data originated from Fitbit users and contains measurements recorded by Fitbit devices. However, the dataset used for this analysis was obtained through Kaggle and provided by Mobius rather than directly from Fitbit. Therefore, the data should be considered a secondary source for this analysis, even though it is the dataset provided for use in the Bellabeat case study.

Comprehensive:
The dataset includes multiple aspects of user wellness, including physical activity, step counts, sleep, heart rate, and weight information. These variables provide sufficient information to examine several behavioral patterns relevant to Bellabeat. However, participant coverage varies across the available tables, and the dataset contains limited demographic information. As a result, the analysis cannot reliably examine how smart-device usage differs across demographic groups such as age or other customer segments.

Current:
The dataset contains Fitbit activity data collected in 2016. Although the data is historical, it remains suitable for this case study because it provides detailed records of fundamental wearable-device behaviors such as steps, activity intensity, sedentary time, and sleep. More recent data could be used in future analysis to determine whether these behavioral patterns have changed over time.

Cited:
The dataset is publicly available through Kaggle and is provided under the CC0: Public Domain license. It is also the dataset identified for use in the Bellabeat case study, providing a clear and accessible source for the analysis.

Licensing, Privacy, Accessibility, and Limitations

The Fitbit Fitness Tracker dataset is distributed under the **CC0: Public Domain** license, allowing it to be freely used for educational and analytical purposes. The dataset contains anonymized user identifiers rather than personally identifiable information, helping protect participant privacy. Because the dataset is publicly available through Kaggle, it is easily accessible for analysis.

Several limitations should also be considered. The number of participants varied across the files used in the analysis. The two daily activity tables contained 35 and 33 distinct participants, the two hourly activity tables contained 34 and 33 participants, and the sleep dataset contained 24 participants. This relatively small and inconsistent participant coverage limits the ability to generalize the findings to the broader population. Additionally, the data represents Fitbit users rather than Bellabeat customers, meaning the insights should be used to inform Bellabeat's marketing strategy rather than describe its existing customer base directly. The Bellabeat case study also notes that the dataset has limitations and recommends considering additional data sources if a more comprehensive analysis is required. Sedentary minutes represent Fitbit's recorded inactivity metric and may be influenced by device wear time and other measurement limitations. Therefore, they should be interpreted as an indicator of inactivity rather than a direct measurement of sitting time.

Data Integrity

Before beginning the analysis, the dataset was reviewed to ensure its integrity. This process included checking for missing values, duplicate records, inconsistent formatting, incorrect data types, and invalid observations. Since multiple CSV files share a common user identifier, the relationships between datasets were also verified before performing joins or transformations. These validation steps helped ensure that the data was accurate, consistent, and suitable for analysis.

Prepare Phase Summary

The Fitbit Fitness Tracker dataset provides an appropriate foundation for analyzing consumer smart device usage. Despite limitations related to sample size and representativeness, the dataset contains sufficient activity and wellness information to identify meaningful usage trends and support the objectives of the Bellabeat case study.

4. Process

Data Processing and Cleaning

The Fitbit data selected during the Prepare phase was imported into Google BigQuery for processing, cleaning, and validation. The original CSV files were maintained as separate raw tables so that the source data would remain unchanged while data-quality issues were investigated. The two available collection periods were initially kept separate to allow their record counts, participant counts, schemas, and overall data quality to be evaluated before determining whether the datasets should be combined for analysis.

The following tables were imported into the bellabeat_case_study dataset:

Table	Records	Unique Users
daily_activity_0312_0411	457	35
daily_activity_0412_0512	940	33
hourly_steps_0312_0411	24,084	34
hourly_steps_0412_0512	22,099	33
sleep_day	413	24

The initial record and participant counts showed that the amount of available data varies across tables and collection periods. The hourly step datasets contain substantially more observations because activity is recorded at an hourly level rather than once per day. The sleep dataset contains data for only 24 unique users, compared with 33 users in the April 12–May 12 daily activity dataset. Therefore, analyses involving both sleep and activity will represent a smaller subset of the Fitbit participants. This difference will be considered when interpreting the results.

Data Type and Import Review

The imported tables were reviewed to confirm that their fields and data types were appropriate for analysis. The daily activity data imported successfully with fields such as ActivityDate stored as a date and numerical activity measurements stored as integer or floating-point values.

The hourly step datasets required additional processing during import. BigQuery initially attempted to interpret the ActivityHour field as a timestamp. However, the original CSV files contained date-time values using an AM/PM format that produced parsing errors. To preserve the original values without losing records, ActivityHour was manually imported as a string. The Id and StepTotal fields were imported as integers. The first CSV row was also identified as a header and excluded from the data during import.

A similar issue occurred with the SleepDay field in the sleep dataset. The field was therefore imported as a string rather than allowing BigQuery to incorrectly interpret the AM/PM portion of the value as a time zone. These date-time fields can subsequently be standardized into appropriate date or datetime formats during processing.

Duplicate Record Check

All five raw tables were checked for exact duplicate records. The duplicate analysis produced the following results:

Table	Total Records	Distinct Records	Duplicate Records
daily_activity_0312_0411	457	457	0
daily_activity_0412_0512	940	940	0
hourly_steps_0312_0411	24,084	24,084	0
hourly_steps_0412_0512	22,099	22,099	0
sleep_day	413	410	3

No exact duplicates were identified in either daily activity or hourly steps dataset. However, three exact duplicate records were identified in the sleep dataset.

The duplicate records corresponded to the following participant-date combinations:

- User 4388161847 on May 5, 2016
- User 4702921684 on May 7, 2016
- User 8378563200 on April 25, 2016

Each of these observations occurred twice in the original sleep dataset. Because the records were exact duplicates, retaining both copies would count the same sleep observation more than once and could distort later calculations involving sleep duration and time in bed.

Duplicate Removal and Validation

Rather than modifying the original raw sleep table, a separate cleaned table named sleep_day_clean was created. Exact duplicate records were removed using SELECT DISTINCT, allowing one valid copy of each duplicated observation to remain while preserving the original sleep_day table.

After cleaning, the new table was validated by comparing its total number of records with its number of distinct records. The validation produced:

- Original sleep records: 413
- Duplicate records removed: 3
- Cleaned sleep records: 410
- Distinct cleaned records: 410
- Remaining exact duplicates: 0

The validation confirmed that the duplicate-removal procedure was successful and that sleep_day_clean contains 410 unique sleep observations.

Missing Values

The five datasets selected for analysis were evaluated for missing values using SQL COUNTIF() queries in BigQuery. The cleaned sleep dataset contained no NULL values across Id, SleepDay, TotalSleepRecords, TotalMinutesAsleep, or TotalTimeInBed. Both hourly steps datasets contained no NULL values across Id, ActivityHour, or StepTotal. The two daily activity datasets were also checked across their activity, distance, active-minute, sedentary-minute, and calorie fields, with no missing values identified. Therefore, no records required removal or imputation because of missing data.

Date Range Validation

The date ranges of the two daily activity datasets were validated in BigQuery using the MIN() and MAX() functions on the ActivityDate field. The first dataset contained records from March 12, 2016 through April 12, 2016, while the second dataset contained records from April 12, 2016 through May 12, 2016. This revealed an overlap on April 12, 2016 between the two collection periods.

Combining the datasets without evaluating this overlap could result in duplicate observations. Therefore, the overlapping records required further examination before the datasets were merged for analysis.

Overlapping Date Investigation and Resolution

Date-range validation identified an overlap on April 12, 2016 between the two daily activity datasets. The earlier dataset contained 24 observations from 24 unique participants on April 12, while the later dataset contained 33 observations from 33 unique participants. An inner join using participant Id and ActivityDate showed that all 24 participants from the earlier dataset also appeared in the later dataset on the overlapping date.

The 24 matching participant-date observations were then compared across all daily activity measurements, including steps, distances, active minutes, sedentary minutes, and calories. None of the 24 matching observations were completely identical; all 24 contained differences in one or more activity measurements. Therefore, these records were not classified as exact duplicates.

To prevent multiple observations for the same participant and date when combining the two collection periods, the April 12 observations from the later dataset were selected for the combined dataset. The later dataset contained more complete participant coverage on that date, with 33 participants compared with 24 in the earlier dataset. The earlier dataset therefore contributed observations through April 11, 2016, while the later dataset contributed observations beginning April 12, 2016.

Daily Activity Dataset Integration and Validation

After resolving the April 12 overlap, the two daily activity tables were combined into a single cleaned table named daily_activity_clean. Records from March 12 through April 11 were retained from the earlier dataset, while records from April 12 through May 12 were retained from the later dataset. The tables were combined using UNION ALL, resulting in 1,373 daily activity records covering March 12 through May 12, 2016.

The combined table was then validated for duplicate records and date coverage. All 1,373 rows were distinct, resulting in zero duplicate records. The final date range extended from March 12 through May 12, 2016. This cleaned table served as the primary daily activity dataset for subsequent analysis and visualization.

Hourly Date-Time Validation

The ActivityHour field in both hourly steps datasets was initially imported as a string because the original AM/PM date-time format was not automatically interpreted correctly by BigQuery during the CSV import process. Before converting the field, SAFE.PARSE_DATETIME() was used to test whether every value could be parsed using the original format (MM/DD/YYYY HH:MM:SS AM/PM). All 24,084 records in the first hourly steps dataset and all 22,099 records in the second dataset were successfully parsed, with zero failed conversions. Therefore, the ActivityHour values can safely be converted to the DATETIME data type for subsequent analysis.

Hourly Steps Dataset Integration and Validation

After validating the ActivityHour field, the date ranges of the two hourly steps datasets were examined before combining them. The first dataset covered March 12 through April 12, 2016, while the second dataset covered April 12 through May 12, 2016. This identified April 12 as an overlapping date between the two collection periods.

Further investigation showed that the first dataset contained 175 hourly observations from 20 participants on April 12, while the second dataset contained 792 observations from 33 participants. Both datasets contained observations throughout the full day, from 12:00 AM through 11:00 PM.

The overlapping records were compared using participant ID and activity hour. All 175 records from the earlier dataset had a matching participant and hour in the later dataset. The StepTotal values were identical for all 175 matching records, with zero differences identified. Therefore, these records were confirmed as duplicates.

To prevent duplicate observations in the combined dataset, April 12 records from the earlier dataset were excluded, while all records from the later dataset were retained. The two datasets were then combined into a cleaned table named hourly_steps_clean, and ActivityHour was converted from STRING to DATETIME.

A final validation of the cleaned table identified 46,008 total records and 46,008 distinct records, resulting in zero duplicate rows. No null values were found in Id, ActivityHour, or StepTotal. The final hourly steps dataset covers March 12, 2016 through May 12, 2016, confirming that the data was successfully combined and prepared for analysis.

5. Analyze

Analysis Approach

After cleaning and validating the Fitbit datasets, I used SQL in BigQuery to analyze trends in daily activity, hourly activity, calories, activity intensity, sleep, and user behavior. I also joined activity and sleep data by participant and date to investigate relationships between physical activity and sleep.

The analysis focused on the following questions:

- When are users most and least active?
- How does physical activity relate to calories burned?
- How do sleep patterns differ among users?
- Is there a relationship between activity and sleep?
- Can users be grouped into meaningful activity segments?

Overall Activity

Across the daily activity dataset, users averaged approximately 7,377 steps per day, 5.29 miles of distance, and 2,295 calories per day.

This establishes the general activity level of the participants before examining more specific behavioral patterns.

Activity by Day of Week

Activity varied throughout the week. Saturday recorded the highest average number of steps at approximately 7,752, while Sunday recorded the lowest at approximately 6,607.

This suggests that user activity patterns vary by day and may provide opportunities for Bellabeat to encourage additional movement on lower-activity days.

Activity Throughout the Day

Hourly step data showed that activity was lowest overnight, increased during the morning, and was considerably higher during daytime and early-evening hours.

For example, average hourly steps increased from only 43 at 12 AM to over 500 during portions of the afternoon and evening. Specifically, average steps reached 534 at 12 PM and rose again during the early evening, reaching the highest hourly average of approximately 555 steps at 7 PM.

Activity and Calories

Daily steps had a moderate positive correlation with calories burned ($r = 0.58$). This means that higher daily step counts generally tended to be associated with higher calorie expenditure. Correlation values range from -1 to +1, with values closer to +1 indicating a stronger positive relationship. Correlation indicates an association between variables and does not establish that one variable causes the other.

When individual activity intensities were compared with calories, very-active minutes had the strongest positive correlation ($r = 0.594$). Fairly-active and lightly-active minutes had weaker positive relationships, while sedentary minutes had a slight negative relationship.

This suggests that higher-intensity physical activity was more strongly associated with calorie expenditure than lower-intensity activity in this dataset.

Sleep Behavior

Among participants with sleep records, average sleep duration was approximately 6.99 hours per night, while average time in bed was approximately 7.64 hours.

Average sleep efficiency was approximately 91.65%, indicating that participants generally spent a high proportion of their time in bed asleep.

Relationship Between Activity and Sleep

After joining daily activity and sleep records by participant and date, 410 matched records from 24 participants were available for combined analysis.

The relationship between daily steps and sleep duration was weak and negative ($r = -0.19$). The Tableau trend analysis indicated that this relationship was statistically significant ($p \approx 0.0001$); however, the small correlation coefficient indicates that the relationship itself was weak. These results show an association within the analyzed sample and should not be interpreted as evidence that increased daily activity causes reduced sleep duration.

User Activity Segmentation

Participants were grouped according to their average daily steps. Users averaging fewer than 5,000 steps per day were classified as Sedentary, 5,000–7,499 steps as Low Active, 7,500–9,999 steps as Somewhat Active, and 10,000 or more steps as Active.

Segment	Users
Sedentary	11
Low Active	9

Somewhat Active	8
Active	7

Of the 35 analyzed participants, 20 were classified as either Sedentary or Low Active - approximately 57%.

The Sedentary group averaged approximately 2,898 steps per day, compared with 12,694 steps per day among Active users.

This segmentation reveals substantial differences in activity behavior among participants and identifies distinct user groups that may benefit from different levels of activity support and engagement.

Data Considerations

Finally, participant engagement was not uniform. Some participants contributed substantially more recorded days than others.

Additionally, the analysis included a relatively small number of users, and only 24 participants had matching activity and sleep records.

Therefore, the results should be interpreted as behavioral trends within this sample, rather than as representative of all wearable-device users or all Bellabeat customers.

6. Share

The results of the analysis were visualized in Tableau to communicate the major patterns identified in Fitbit user activity, calories burned, sedentary behavior, and sleep. Two dashboards were created to organize the findings and make the results easier to interpret.

Tableau Dashboards

Bellabeat Wellness & Activity Analysis

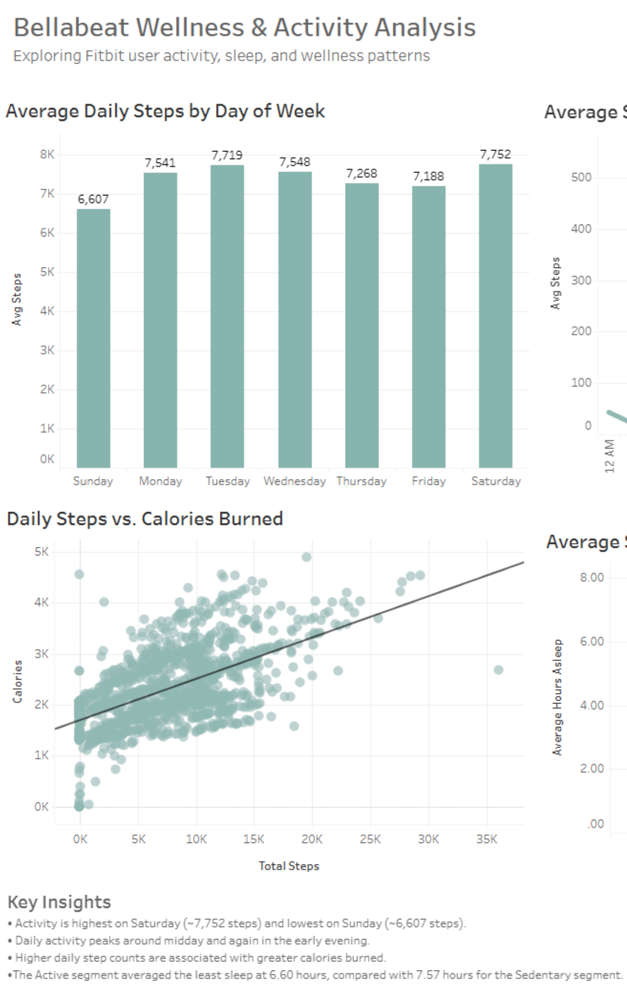

Figure 1. Bellabeat Wellness & Activity Analysis Dashboard

Interactive Dashboard: [View on Tableau Public]

The first dashboard provides an overview of user activity and its relationship with calories burned and sleep duration. It contains four visualizations:

Average Daily Steps by Day of Week
This bar chart compares average daily step counts across the week. Saturday recorded the highest average at approximately 7,752 steps, while Sunday recorded the lowest at approximately 6,607 steps. Overall, users tended to be more active on Saturdays and less active on Sundays.

Average Steps by Hour of Day
This line chart shows how user activity changes throughout the day. Activity is very low during overnight and early-morning hours before increasing rapidly beginning around 6 AM. Activity remains relatively high throughout the daytime, with noticeable peaks around midday and 6–7 PM.

Daily Steps vs. Calories Burned
This scatterplot examines the relationship between total daily steps and calories burned. The upward-sloping trend line indicates a positive relationship, with higher daily step counts generally associated with greater calories burned.

Average Sleep Duration by Activity Level
This bar chart compares average sleep duration across sedentary, low-active, somewhat-active, and active users. The Sedentary group recorded the highest average sleep duration at approximately 7.57 hours, while the Active group recorded the lowest at approximately 6.60 hours. Although sleep duration did not decrease consistently across every activity category, the results show differences in average sleep duration among the four activity segments.

Bellabeat Sleep & Activity Analysis

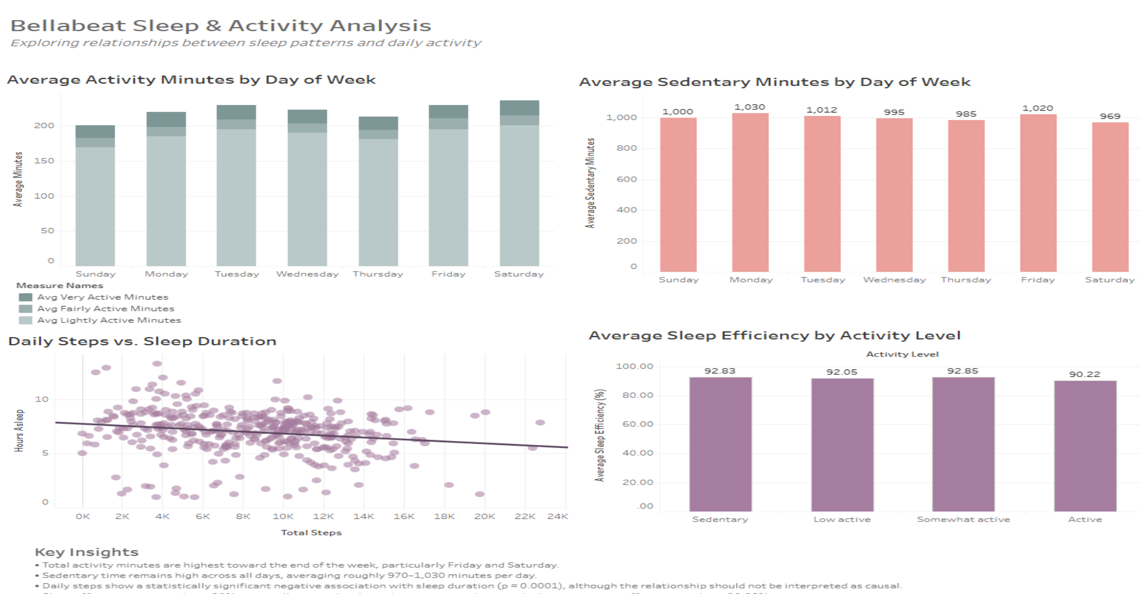

Figure 2. Bellabeat Sleep & Activity Analysis Dashboard

Interactive Dashboard: View on Tableau Public

The second dashboard focuses more closely on the relationships between physical activity, sedentary behavior, and sleep. It contains four visualizations:

Average Activity Minutes by Day of Week
This stacked bar chart compares lightly active, fairly active, and very active minutes throughout the week. Total activity minutes were generally highest toward the end of the week, particularly on Friday and Saturday.

Average Sedentary Minutes by Day of Week
This bar chart compares average recorded sedentary time across the week. Sedentary minutes remained high on every day, averaging approximately 970–1,030 minutes per day. Monday had the highest average at approximately 1,030 minutes, while Saturday had the lowest at approximately 969 minutes.

Daily Steps vs. Sleep Duration
This scatterplot examines the relationship between total daily steps and hours asleep. The downward trend line indicates a negative association between the two variables. The relationship was statistically significant ($p \approx 0.0001$); however, the analysis demonstrates an association and should not be interpreted as evidence that increased activity causes reduced sleep.

Average Sleep Efficiency by Activity Level
This bar chart compares sleep efficiency across activity levels. Average sleep efficiency remained above 90% for all four groups. Sedentary, low-active, and somewhat-active users averaged approximately 92–93%, while active users had the lowest average sleep efficiency at approximately 90.22%.

Key Findings

The Tableau analysis revealed several important patterns in Fitbit user behavior:

- Saturday was the most active day by average steps, with approximately 7,752 steps, while Sunday was the least active at approximately 6,607 steps.
- User activity was lowest overnight, increased considerably during the morning, and showed notable activity peaks around midday and the early evening.
- Higher daily step counts were associated with greater calories burned, indicating a clear positive relationship between movement and energy expenditure.
- Recorded sedentary minutes remained high throughout the week, averaging approximately 970–1,030 minutes per day.

- Higher daily step counts showed a statistically significant negative association with sleep duration, although this relationship should not be interpreted as causal.
- The Active group recorded the lowest average sleep duration at approximately 6.60 hours, while the Sedentary group recorded the highest at approximately 7.57 hours.
- Sleep efficiency remained relatively high across all activity levels at 90% or greater, although the active group had the lowest average sleep efficiency at approximately 90.22%.

Overall, the visualizations show that Fitbit users' activity varies by both day of the week and time of day, while also revealing relationships among physical activity, calories burned, sedentary behavior, and sleep. These findings provide the foundation for developing actionable recommendations for Bellabeat in the Act phase of the analysis.

7. Act

The final phase of the analysis focuses on translating the findings into actionable recommendations for Bellabeat. The analysis identified several patterns in Fitbit user behavior related to daily activity, sedentary time, calories burned, and sleep. These findings can be used to inform Bellabeat's marketing strategy and demonstrate how its wellness products can help consumers better understand their daily habits.

Selected Bellabeat Product: Leaf

The Bellabeat Leaf was selected as the product to which the findings of this analysis were applied. Leaf is a wellness tracker that can be worn as a bracelet, necklace, or clip and connects with the Bellabeat app to track activity, sleep, and stress.

Leaf is particularly relevant to this analysis because the Fitbit dataset revealed meaningful patterns involving physical activity and sleep. Positioning Leaf as a tool that helps users understand the relationship between these behaviors could allow Bellabeat to communicate the value of the product through practical, data-driven wellness insights.

Marketing Recommendations

1. Promote Leaf as a Tool for Building Consistent Activity Habits

Bellabeat could market Leaf as a tool that helps users recognize periods of lower activity and develop more consistent movement habits. The analysis showed that average daily steps varied throughout the week, with Saturday recording the highest average at approximately 7,752 steps and Sunday recording the lowest at approximately 6,607 steps. Hourly activity also varied throughout the day, with activity increasing during the morning and reaching its highest levels during the daytime and early evening.

Bellabeat could use these behavioral patterns in marketing campaigns that emphasize Leaf's ability to track daily movement and help users become more aware of when they are most and least active. Marketing messages could encourage consumers to use their activity data to establish personalized goals, participate in weekend challenges, and maintain consistent movement throughout the week.

2. Emphasize Reducing Prolonged Sedentary Behavior

Bellabeat could use Leaf to help users become more aware of prolonged periods of inactivity and encourage movement throughout the day through personalized reminders and activity goals. The analysis found that users averaged approximately 970–1,030 recorded sedentary minutes per day, indicating that recorded inactivity was substantial throughout the dataset.

Marketing campaigns could emphasize how tracking everyday movement with Leaf can help users become more aware of their sedentary habits. Rather than focusing exclusively on achieving a specific daily step count, Bellabeat could promote a broader message of incorporating movement throughout the day. This could position Leaf as a product designed to support sustainable daily wellness habits rather than only formal exercise.

3. Market Leaf as an Integrated Activity and Sleep Wellness Tracker

Bellabeat could differentiate Leaf by emphasizing its ability to track multiple aspects of wellness within a single product. The analysis identified relationships between physical activity and sleep behavior. The Active group recorded the lowest average sleep duration at approximately 6.60 hours, while the Sedentary group recorded the highest at approximately 7.57 hours. Daily steps also showed a statistically significant negative association with sleep duration in this dataset, although this relationship should not be interpreted as evidence that increased activity causes reduced sleep.

Sleep efficiency remained above 90% across all activity levels, showing that activity level and sleep can provide different but complementary information about user behavior. Bellabeat could therefore market Leaf as a tool that helps consumers view activity and sleep together rather than treating each metric independently. Marketing messages could emphasize understanding overall wellness patterns, balancing movement with rest, and using personal data to develop more informed daily routines.

Future Improvements and Additional Data

Although the Fitbit dataset provided useful insights into smart-device usage, several limitations should be considered when applying the results to Bellabeat's marketing strategy. Future analysis would benefit from a larger and more diverse sample of users collected over a longer period of time. This would help determine whether the patterns identified in this analysis remain consistent across different populations and seasons.

Additional demographic information such as age and other relevant user characteristics could also help Bellabeat determine whether activity and sleep patterns differ among customer segments. The dataset used for this analysis did not provide sufficient demographic information to conduct this type of segmentation.

Future analysis could also incorporate additional wellness measures that are relevant to Bellabeat products, such as stress and other available wellness behaviors. Because Leaf tracks activity, sleep, and stress, analyzing these measures together could provide a more complete understanding of how consumers use wellness technology.

Finally, analyzing Bellabeat's own customer data, when appropriately collected and handled, could provide insights that are more directly representative of Bellabeat's target customers than third-party Fitbit data. This could allow future recommendations to be based on actual Bellabeat product usage and customer behavior.

Conclusion

The analysis identified several trends in smart-device usage that could help inform Bellabeat's marketing strategy. User activity varied by both day of the week and time of day, higher daily step counts were associated with greater calories burned, and recorded sedentary minutes remained substantial throughout the week. The analysis also identified relationships between activity and sleep behavior.

Based on these findings, Bellabeat could market Leaf as a wellness tracker that helps consumers develop more consistent activity habits, become more aware of prolonged sedentary behavior, and understand activity and sleep as interconnected components of overall wellness. These recommendations use the behavioral patterns identified in the Fitbit dataset to demonstrate how Bellabeat could communicate the value of Leaf to potential customers.

8. References

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